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AI-Native Ecosystem — Comprehensive Implementation Guide

Lagos, Nigeria — pechgroupholdings.tech

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1. EXECUTIVE SUMMARY

PECH Group Holdings Ltd is building an **AI-native digital infrastructure platform** for Africa, starting from Lagos, Nigeria. The ecosystem connects **10 business verticals** through a **unified AI core** — similar in architecture to platforms built by Alibaba Group, Amazon, and Tencent, but optimized for African markets, energy infrastructure, and commerce.

Core principles: - **Self-hosted AI** — no vendor lock-in, all models run on our own servers - **Open-source only** — Apache-2.0 and MIT licensed tools that we can rebrand, modify, and close-source - **RAG-first** — models learn from business data daily without expensive training - **Nigeria-optimized** — power backup, import-duty-aware budgets, local infrastructure - **Phase-based deployment** — start lean, scale with revenue

The platform comprises ~78 microservices across 12 domains, powered by ~20-25 open-source systems and 15+ AI/ML models, serving customers, merchants, installers, agents, and developers through a unified identity and data layer.

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2. ECOSYSTEM OVERVIEW

The 10 Interconnected Business Verticals

```
                    PECH AI CORE
                        |
   +------------------+------------------+
   |        |         |         |        |
 Solar   Market-   Fintech    PMAP   Logistics
 Energy  place     Wallet    Mapping  Network
   |        |         |         |        |
```

```
    +-----------------+-----------------+
    |        |        |        |        |
   IoT   Developer  Installer  ERP   Real Estate
 Devices  Platform  Ecosystem System  Platform
```

#	Vertical	What It Does	AI Role
1	**Solar Intelligence**	Panel, inverter, battery distribution	System sizing, customer advisory, troubleshooting
2	**Marketplace**	Multi-vendor product commerce	Search AI, recommendations, product descriptions, seller analytics
3	**Fintech**	Wallet, payments, agents, escrow	Fraud detection, transaction classification, spending insights
4	**PMAP / AMAP**	Digital mapping for African markets	Business discovery, route optimization, location ads
5	**Logistics**	Hub-and-spoke distribution network	Demand forecasting, route optimization, warehouse management
6	**IoT Devices**	Smart switches, locks, meters, cameras	Voice control, automation, device troubleshooting
7	**Developer Platform**	Open APIs + AI coding assistant	Code generation, API docs, SDK, developer onboarding
8	**Installer Ecosystem**	Certified installation network	Installation guidance, troubleshooting, training, certification
9	**ERP System**	Accounting, inventory, HR, procurement	Financial reports, inventory prediction, tax compliance, auditing
10	**Real Estate**	Property buying, selling, renting	Property recommendations, price prediction, virtual tours

Data Flow: Every Action Feeds the AI

```
 Customer buys solar panel
        |
Marketplace records transaction
        |
Logistics schedules delivery
        |
Installer receives job
        |
Fintech processes payment
        |
AI analyzes demand patterns
        |
System becomes smarter
```

All verticals share a **central data lake** and **unified AI core**. This creates powerful network effects — the more data flows, the smarter every vertical becomes.

3. AI MODEL STACK — FACT-CHECKED & LICENSE-VERIFIED

Core LLM Models (All Apache-2.0 or MIT — can rebrand and close-source)

Model	Org	Size	License	VRAM (FP16)	VRAM (4-bit)	Best For	Download
Qwen2.5-7B	Alibaba	7B	Apache-2.0	~14GB	~4-6GB	General reasoning, central AI assistant	https://huggingface.co/Qwen
Qwen2.5-0.5B	Alibaba	0.5B	Apache-2.0	~1GB	~0.5GB	Edge AI, IoT devices, lightweight tasks	https://huggingface.co/Qwen
Qwen2.5-1.5B	Alibaba	1.5B	Apache-2.0	~3GB	~1.5GB	Small chatbots, customer support	https://huggingface.co/Qwen
Qwen-VL	Alibaba	7B	Apache-2.0	~14GB	~4-6GB	Image analysis, document reading, receipts	https://huggingface.co/Qwen
Phi-3 Mini	Microsoft	3.8B	MIT	~8GB	~3GB	Reasoning, analytics, recommendations	https://huggingface.co/microsoft
Phi-4	Microsoft	14B	MIT	~28GB	~8-10GB	Advanced reasoning (cloud/production only)	https://huggingface.co/microsoft
Mistral-7B	Mistral AI	7B	Apache-	~14GB	~4-6GB	Customer service chatbot, content	https://huggingface.co/mistralai

			2.0			generation	
Mixtral-8x7B	Mistral AI	46B MoE	Apache-2.0	~90GB	~24-30GB	Complex reasoning, fraud detection (cloud)	https://huggingface.co/mistralai
DeepSeek-Coder-V2	DeepSeek	6.7B	MIT	~14GB	~4-6GB	Code generation, developer assistant	https://huggingface.co/deepseek-ai
SmolLM	HuggingFace	135M-1.7B	Apache-2.0	~0.3-3.5GB	N/A	Ultra-lightweight support bot, edge AI	https://huggingface.co/HuggingFaceTB
TinyLlama	Community	1.1B	Apache-2.0	~2.5GB	~1GB	Simple chatbot, IoT assistants	https://github.com/jzhang38/TinyLlama
Whisper Large-v3	OpenAI	1.5B	MIT	~3GB	N/A	Speech-to-text, voice commands	https://github.com/openai/whisper

Specialized ML Models (Classification, Prediction, Detection)

Model	License	Size	Best For	Alternative	Recommended	Why
BERT	Apache-2.0	110M	Intent classification, text classification	DistilBERT (66M)	**DistilBERT**	97% of BERT accuracy at 40% the size and 60% faster
DistilBERT	Apache-2.0	66M	Sentiment analysis, expense categorization	BERT (110M)	**DistilBERT**	Lighter, faster, sufficient for PECH tasks
MiniLM	MIT	22M	Embeddings for search and similarity	E5-Small (33M)	**MiniLM**	Smaller, faster inference, excellent quality
E5-Small	MIT	33M	Search embeddings, semantic search	MiniLM (22M)	MiniLM (unless multilingual needed)	E5 better for multilingual
XGBoost	Apache-2.0	N/A	Fraud detection, credit scoring	LightGBM (MIT)	**XGBoost** for fraud, **LightGBM** for forecasting	XGBoost more robust for imbalanced data; LightGBM faster training
LightGBM	MIT	N/A	Demand forecasting, price prediction	XGBoost (Apache-2.0)	**LightGBM** for time-series	Faster, handles large datasets well
Prophet	MIT	N/A	Time-series forecasting (demand, revenue)	StatsForecast (Apache-2.0)	**Prophet** for simplicity	Easy API; StatsForecast for when you need multiple algorithms
LightFM	Apache-2.0	N/A	Hybrid recommendations	RecBole (MIT)	**LightFM** for production	Battle-tested; RecBole is better for research/experimentation
Isolation Forest	BSD (scikit-learn)	N/A	Anomaly detection (auditing)	PyOD (BSD)	**Isolation Forest**	Built into scikit-learn, no extra dependency

Speech & Voice Models

Model	License	Best For	Alternative	Recommended
Whisper (OpenAI)	MIT	Speech-to-text	whisper.cpp (MIT), Kaldi (Apache-2.0)	**Whisper** — best accuracy; use **whisper.cpp** for edge/CPU
Piper TTS	MIT	Text-to-speech	Kokoro TTS (Apache-2.0), MeloTTS (MIT)	**Piper** — fastest, smallest, great quality
Kokoro TTS	Apache-2.0	High-quality TTS	Piper (MIT)	Use when quality matters more than speed

Vision & Document Processing

Model	License	Best For	Alternative	Recommended
PaddleOCR	Apache-2.0	Invoice/receipt OCR	Tesseract (Apache-2.0), docTR (Apache-2.0)	**PaddleOCR** — best accuracy for structured docs
Tesseract	Apache-2.0	General OCR	PaddleOCR	Use for simple text extraction
RT-DETR	Apache-2.0	Object detection	Detectron2 (Apache-2.0)	**RT-DETR** — replaced YOLOv8 (AGPL license issue)

Detectron2	Apache-2.0	Instance segmentation	RT-DETR	Use when you need segmentation, not just detection
OpenCV	Apache-2.0	Image processing	N/A	**Essential** — used in every vision pipeline
Qwen-VL	Apache-2.0	Image understanding, photo analysis	N/A	**Qwen-VL** — only multimodal LLM in our stack

Translation

Model	License	Best For	Alternative	Recommended
Argos Translate	MIT	Offline translation	OpenNMT (MIT), MarianMT (MIT)	**Argos** — easiest to deploy, offline capable
MarianMT	MIT	Batch translation	Argos Translate	Use for high-volume translation pipelines

Image Editing AI (User-Facing Marketplace Features)

Model/Tool	License	Capability	Notes
rembg (U2-Net)	MIT	Background removal	Lightweight, fast, CPU-capable
Stable Diffusion XL	CreativeML Open RAIL-M	Image generation, virtual staging	**Note:** Open RAIL license has restrictions on harmful content but allows commercial use and modification
InstructPix2Pix	MIT	Image editing via text instructions	"Make this brighter", "remove the watermark"
OpenCV	Apache-2.0	Basic image editing (resize, crop, filters)	Foundation layer for all image features

4. MODEL-TO-BUSINESS MAPPING

Which Models Power Which Vertical

Vertical	Primary Model	Secondary Model	ML Models	Speech/Vision
Solar	Qwen2.5-7B (sizing, advisory)	Phi-3 Mini (recommendations)	Prophet (demand forecast)	Whisper (voice)
Marketplace	Mistral-7B (product descriptions)	Phi-3 Mini (search ranking)	LightFM (recommendations), LightGBM (pricing)	rembg (product photos)
Fintech	Qwen2.5-7B (support, insights)	Mixtral-8x7B (fraud, cloud)	XGBoost (fraud detection), Isolation Forest (anomalies)	PaddleOCR (receipts)
PMAP	Qwen2.5-7B (business search)	Phi-3 Mini (location ranking)	LightGBM (demand mapping)	—
Logistics	Qwen2.5-7B (route optimization)	Phi-3 Mini (demand forecast)	Prophet (inventory prediction)	Whisper (driver voice)
IoT	SmolLM / TinyLlama (edge)	Qwen2.5-0.5B (device assistant)	—	Whisper (voice control)
Developer Platform	DeepSeek-Coder (code gen)	Qwen2.5-7B (docs assistant)	—	—
Installer	Qwen2.5-7B (technical assistant)	Mistral-7B (troubleshooting)	—	Whisper (voice), Qwen-VL (photo analysis)
ERP	Phi-3 Mini (reports, analytics)	Qwen2.5-7B (accounting AI)	XGBoost (anomaly), LightGBM (forecast)	PaddleOCR (invoices)
Real Estate	Qwen2.5-7B (property assistant)	Phi-3 Mini (price prediction)	LightFM (recommendations), LightGBM (valuation)	Qwen-VL (property photos)

5. OPEN-SOURCE PLATFORM STACK

All tools are verified **Apache-2.0 or MIT** unless noted. Tools with license issues have been **replaced**.

#	Category	Tool	License	Replaces	Link

1	**Identity & Auth**	Keycloak	Apache-2.0	—	https://github.com/keycloak/keycloak
2	**API Gateway**	Apache APISIX	Apache-2.0	—	https://github.com/apache/apisix
3	**ERP System**	ERPNext	MIT	—	https://github.com/frappe/erpnext
4	**Core Banking**	Apache Fineract	Apache-2.0	—	https://github.com/apache/fineract
5	**Marketplace**	Medusa	MIT	—	https://github.com/medusajs/medusa
6	**Logistics Routing**	OSRM	BSD	—	https://github.com/Project-OSRM/osrm-backend
7	**Real Estate**	Medusa (extended)	MIT	Sharetribe Flex (proprietary)	Extend Medusa with property listing modules
8	**Procurement**	OpenProcurement	Apache-2.0	—	https://github.com/openprocurement
9	**IoT Platform**	ThingsBoard CE	Apache-2.0	—	https://github.com/thingsboard/thingsboard
10	**Customer Support**	Chatwoot	MIT	—	https://github.com/chatwoot/chatwoot
11	**Workflow Automation**	Apache Airflow	Apache-2.0	n8n (Sustainable Use License)	https://github.com/apache/airflow
12	**Analytics/BI**	Apache Superset	Apache-2.0	Metabase (AGPL)	https://github.com/apache/superset
13	**AI Orchestration**	LangChain	MIT	—	https://github.com/langchain-ai/langchain
14	**Agent Framework**	LangGraph	MIT	AutoGen (less mature)	https://github.com/langchain-ai/langgraph
15	**LLM Serving (dev)**	Ollama	MIT	—	https://github.com/ollama/ollama
16	**LLM Serving (prod)**	vLLM	Apache-2.0	—	https://github.com/vllm-project/vllm
17	**Vector Database**	Qdrant	Apache-2.0	—	https://github.com/qdrant/qdrant
18	**Relational DB**	PostgreSQL	PostgreSQL License	—	https://www.postgresql.org
19	**Graph DB**	Apache AGE	Apache-2.0	Neo4j (GPL/Commercial)	https://github.com/apache/age
20	**Analytics DB**	DuckDB	MIT	—	https://github.com/duckdb/duckdb
21	**Cache**	Redis	BSD	—	https://github.com/redis/redis
22	**Event Streaming**	Apache Kafka	Apache-2.0	—	https://github.com/apache/kafka
23	**Object Detection**	RT-DETR	Apache-2.0	YOLOv8 (AGPL)	https://github.com/PaddlePaddle/PaddleDetection
24	**Recommendation**	LightFM	Apache-2.0	—	https://github.com/lyst/lightfm
25	**Fraud Detection**	XGBoost	Apache-2.0	—	https://github.com/dmlc/xgboost
26	**Forecasting**	Prophet	MIT	—	https://github.com/facebook/prophet
27	**OCR**	PaddleOCR	Apache-2.0	—	https://github.com/PaddlePaddle/PaddleOCR
28	**Embeddings**	Sentence Transformers	Apache-2.0	—	https://github.com/UKPLab/sentence-transformers
29	**Monitoring**	Grafana	AGPL-3.0	—	Use for internal monitoring only (AGPL OK for internal)
30	**Container Orchestration**	Kubernetes	Apache-2.0	—	https://github.com/kubernetes/kubernetes

License Replacements Made

Original Tool	Problem	Replacement	Why
Metabase	AGPL-3.0 — must share modified source	**Apache Superset**	Same BI capabilities, Apache-2.0
n8n	Sustainable Use License — can't resell	**Apache Airflow**	Full workflow automation, no restrictions
YOLOv8	AGPL-3.0 — copyleft, must share source	**RT-DETR**	Real-time detection, Apache-2.0, comparable accuracy
Sharetribe Flex	Proprietary SaaS	**Medusa (extended)**	Already in our stack, MIT license
Neo4j	GPL Community / Commercial Enterprise	**Apache AGE**	Runs as PostgreSQL extension, Apache-2.0
AutoGen	Microsoft, less mature	**LangGraph**	More mature agent orchestration, MIT

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6. MICROSERVICE ARCHITECTURE

78 Microservices Across 11 Domains

Domain 1: Identity & Security (6 services)

Service	Purpose	Platform
Auth Service	Login, OAuth2, SSO	Keycloak
User Identity Service	Unified user profiles	Keycloak
Role Management Service	Roles and permissions	Keycloak
Session Service	Session tokens	Keycloak
API Key Service	Developer API authentication	Custom + APISIX
Audit Log Service	Security logs	Custom

Domain 2: User Platform (8 services)

Service	Purpose
Customer Service	Manage end customers
Merchant Service	Manage marketplace vendors
Installer Service	Solar/IoT installers
Agent Service	Fintech agents
Employee Service	Company staff
Profile Service	User profile data
Notification Service	Email, SMS, push
Messaging Service	In-app communication

Domain 3: Marketplace (10 services)

Service	Purpose	Platform
Product Catalog Service	Product listings	Medusa
Category Service	Product categories	Medusa
Inventory Service	Stock management	Medusa + ERPNext
Order Service	Order lifecycle	Medusa
Checkout Service	Checkout flow	Medusa
Pricing Service	Dynamic pricing	Custom + LightGBM
Discount Service	Promotions	Medusa
Review Service	Ratings and reviews	Custom
Merchant Settlement	Vendor payouts	Fineract
Search Service	AI-powered search	Qdrant + MiniLM

Domain 4: Solar Energy (7 services)

Service	Purpose
Solar Product Service	Panels, inverters, batteries catalog

Installation Service	Track installations
Installer Matching Service	Assign nearest qualified installer
Energy Monitoring Service	Monitor solar generation
Warranty Service	Equipment warranties
Maintenance Service	Service scheduling
Solar Analytics Service	Energy production analytics

Domain 5: Fintech (9 services)

Service	Purpose	Platform
Wallet Service	User wallets	Fineract
Payment Service	Payment processing	Fineract + Paystack/Flutterwave
Transaction Ledger	Financial records	Fineract
Agent Management	Fintech field agents	Custom
Commission Service	Agent commissions	Custom
Escrow Service	Marketplace escrow	Fineract
Fraud Detection Service	Detect fraud	XGBoost + Mixtral
Loan Service	Microloans	Fineract
Compliance Service	CBN regulatory checks	Custom

Domain 6: Logistics (7 services)

Service	Purpose	Platform
Shipment Service	Shipment creation	Custom
Route Optimization	Delivery routes	OSRM
Fleet Management	Vehicle tracking	Custom + GPS
Warehouse Service	Inventory storage	ERPNext
Tracking Service	Delivery tracking	Custom + Kafka
Delivery Assignment	Assign drivers	Custom + AI
Logistics Analytics	Performance insights	Superset

Domain 7: Real Estate (7 services)

Service	Purpose
Property Listing Service	Buy/sell/rent listings
Property Search Service	AI-powered property search
Property Recommendation	Suggested properties
Rental Management	Rental operations
Property Analytics	Price trends, valuation
Booking Service	Viewing appointments
Agent Management	Property agents

Domain 8: Procurement (4 services)

Service	Purpose	Platform
Tender Listing	Tender opportunities	OpenProcurement
Bid Submission	Bid management	OpenProcurement
Contract Management	Contract lifecycle	OpenProcurement

Supplier Service	Vendor records	ERPNext

Domain 9: IoT (6 services)

Service	Purpose	Platform
Device Registry	IoT device records	ThingsBoard
Telemetry Service	Sensor data	ThingsBoard
Device Command	Remote control	ThingsBoard
Alert Service	System alerts	ThingsBoard
Firmware Update	OTA updates	ThingsBoard
Device Analytics	IoT insights	ThingsBoard + Superset

Domain 10: Customer Support (4 services)

Service	Purpose	Platform
Ticket Service	Support tickets	Chatwoot
Chat Service	Live chat + WhatsApp	Chatwoot
Knowledge Base	Help center	Chatwoot
AI Support Bot	Automated responses	SmolLM + LangChain

Domain 11: AI & Data (10 services)

Service	Purpose	Models Used
Recommendation Engine	Product, property suggestions	LightFM
Dynamic Pricing	Price optimization	LightGBM
Fraud Detection	Payment fraud analysis	XGBoost, Isolation Forest
Customer Segmentation	User clustering	scikit-learn
Search Ranking	Relevance scoring	MiniLM embeddings
Demand Forecasting	Inventory, demand prediction	Prophet, LightGBM
Recruitment AI	Resume parsing, candidate scoring	DistilBERT, MiniLM
Accounting AI	Expense classification, reports	BERT, Phi-3
Tax AI	VAT calculations, compliance	SmolLM, rules engine
Audit AI	Anomaly detection	Isolation Forest, Z-score

7. USER-FACING AI FEATURES & SUBSCRIPTION TIERS

Marketplace AI Features for Users

Feature	Description	AI Model(s)	Tier
Product photo background removal	Remove/replace product backgrounds	rembg (U2-Net)	Free
Smart product title generation	AI writes product titles from photos/description	Qwen2.5-7B	Free
Basic price recommendation	Suggest selling price based on similar products	LightGBM	Free
Product description generation	AI writes full product descriptions	Mistral-7B	Paid Tier 1
Image enhancement	Auto-improve product photo quality	OpenCV	Paid Tier 1
Voice-to-edit	Edit product details via voice commands	Whisper + Qwen2.5	Paid Tier 1
AI business insights	Sales trends, best-selling analysis	Phi-3 Mini + LightGBM	Paid Tier 2
Advanced pricing AI	Dynamic pricing with demand/competition analysis	LightGBM + Qwen2.5	Paid Tier 2
Bulk product editing	AI-assisted batch editing via voice/text	Whisper + Mistral-7B	Paid Tier 2

Competitor analysis	Market positioning and pricing intelligence	Phi-3 Mini	Paid Tier 3
Custom AI assistant	Personalized business AI trained on seller's data	Qwen2.5-7B + RAG	Paid Tier 3

Subscription Tiers

Tier	Name	Price (₦/month)	Key AI Features
Free	Basic Seller	₦0	Background removal, basic title gen, price suggestion
Tier 1	Pro Seller	₦2,000-₦5,000	+ product descriptions, image enhancement, voice-to-edit
Tier 2	Business	₦10,000-₦20,000	+ AI insights, dynamic pricing, bulk editing
Tier 3	Enterprise	₦50,000+	+ competitor analysis, custom AI assistant, priority support

8. ACCOUNTING, AUDITING & TAX AI

AI-Powered Financial Intelligence (ERP Add-on)

This module can be accessed in three ways: 1. **Included in ERP subscription** — certain tiers include basic accounting AI 2. **Add-on for ERP users** — upgrade subscription to add advanced features 3. **Standalone subscription for accountants** — accountants subscribe directly

Accounting AI

Feature	Model	Description
Invoice classification	DistilBERT	Auto-categorize invoices (inventory, shipping, marketing, tax, salary)
Expense categorization	DistilBERT	Classify expenses from receipts/transactions
OCR invoice extraction	PaddleOCR	Extract data from paper invoices and receipts
Financial summaries	Phi-3 Mini	Generate monthly/quarterly financial reports
Anomaly detection	Isolation Forest	Flag unusual transactions
Cash flow forecasting	Prophet	Predict future cash flow based on history

Auditing AI

Feature	Model	Description
Duplicate invoice detection	MiniLM embeddings	Find duplicate or near-duplicate invoices
Abnormal expense flagging	Z-score (Isolation Forest)	Flag transactions where
Revenue spike analysis	LightGBM	Detect unusual revenue patterns
Inventory mismatch	Rules + XGBoost	Compare reported vs actual inventory
Compliance checking	SmolLM + rules	Verify against Nigerian accounting standards

Tax AI

Feature	Model	Description
VAT calculation	Rules engine	7.5% Nigerian VAT auto-calculation
WHT computation	Rules engine	Withholding tax for various categories
Corporate tax estimation	Phi-3 Mini	Annual tax liability projection
Tax report generation	SmolLM	Generate FIRS-compliant reports
Invoice verification	PaddleOCR + rules	Verify tax invoices against regulations

Subscription Model for Accounting AI

Tier	Price (₦/month)	Features
Basic (ERP included)	₦0 (with ERP)	Invoice classification, expense categorization, VAT calc
Professional	₦15,000-₦25,000	+ OCR extraction, anomaly detection, financial summaries

Enterprise	₦50,000-₦100,000	+ full auditing AI, tax automation, custom reports
Accountant License	₦10,000-₦20,000	Standalone for accountants — multi-client management

9. HR & RECRUITMENT AI

AI-Powered Recruitment Pipeline (Cross-Platform)

One recruitment AI engine supports hiring across all verticals:

Platform	Hiring For
Solar company	Installers, technicians
Marketplace	Vendors, merchants
Fintech	Agents
Logistics	Drivers, delivery partners
Real estate	Property agents

Pipeline Architecture

```
Job Posting → Application Form → ATS Database → Resume Parser
     → LLM Skill Extractor → Embedding Engine → Candidate Ranking
     → Interview Generator → Recruitment Dashboard
```

Models Used

Component	Model	License
Resume parsing	PyResparser + DistilBERT	Apache-2.0
Skill extraction	SmolLM	Apache-2.0
Candidate embeddings	MiniLM	MIT
Candidate similarity	FAISS	MIT
Candidate ranking	Custom (weighted scoring)	—

Ranking Algorithm

```
Score = (Skill_Match × 0.50) + (Experience × 0.30) + (Education × 0.20)
```

Free Tools for HR

Tool	Purpose	License
OpenCATS	ATS (applicant tracking)	GPL (internal use OK)
ERPNext HR module	Employee records, payroll	MIT
Google Forms	Application collection	Free
Google Sheets	Candidate database	Free

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10. RAG PIPELINE — HOW MODELS LEARN FROM BUSINESS DATA

Since PECH is **not training models** initially, we use **Retrieval-Augmented Generation (RAG)** — models retrieve relevant business data before generating responses.

RAG Architecture

```
Business Data Sources
    |
    ├── Product catalogs
    ├── Customer transactions
    ├── ERP records
    ├── Solar installation data
    ├── PMAP location data
    ├── Support chat history
    |
    v
Data Cleaning & Processing
    |
    v
Embedding Generation (MiniLM / E5-Small)
    |
    v
Vector Database (Qdrant)
    |
    v
User Query → LLM retrieves relevant docs → AI Response
```

Technology Stack

Component	Tool	License
Embeddings	Sentence Transformers (MiniLM)	MIT
Vector DB	Qdrant	Apache-2.0
Orchestration	LangChain	MIT
Agent framework	LangGraph	MIT
LLM serving (dev)	Ollama	MIT
LLM serving (prod)	vLLM	Apache-2.0

How Data Flows In Daily

1. New products added to marketplace → embedded → stored in Qdrant
2. Customer support chats → logged → embedded → improve future responses
3. Solar installation reports → embedded → installer AI learns from them
4. Financial transactions → analyzed → accounting AI improves
5. PMAP location updates → embedded → search AI improves

The models get smarter every day without any expensive training.

11. HARDWARE INFRASTRUCTURE — CHINA SOURCING & NIGERIA COSTS

Phase 1: TRX50 AI Workstation (Zero-Waste Single Platform)

Component	Specification	China Price
GPU	RTX 4090 24GB	$1,200-$1,400
CPU	Threadripper 7960X (24C/48T, sTR5)	$830-$1,100
Motherboard	ASUS Pro WS TRX50-SAGE WiFi (3× PCIe 5.0 x16, 8 DIMM)	$550-$700
RAM	128GB DDR5-5600 ECC RDIMM (4×32GB)	$280-$400
Storage (OS)	1TB Samsung 990 Pro NVMe	$50-$70
Storage (Data)	4TB Samsung 990 Pro NVMe	$200-$280
PSU	Seasonic PRIME TX-1600 80+ Titanium	$350-$450
Case	4U Rackmount chassis	$120-$180
Cooling	Noctua NH-U14S TR5-SP6	$80-$110
Fans	3× 140mm Noctua NF-A14	$30-$45
Hardware subtotal		**$3,690-$4,735**

Shipping + duties + clearing	~35% markup	
Total landed Lagos		**₦8.2M-₦11M**

Phase 2: Add GPU + RAM + Storage (Upgrade, NOT Replacement)

- 2nd RTX 4090 ($1,200-$1,400) + 128GB ECC RDIMM ($280-$400) + 4TB NVMe ($200-$280)
- **Total landed Lagos:** ₦4.1M-₦5.3M (additions only — every Phase 1 component kept)

RTX 5090 Option (32GB VRAM, future-proof)

- 32GB vs 24GB = run larger models without quantization
- China price: $1,800-$2,200/card
- Substitute for Phase 2 GPU if available at time of upgrade

Where to Buy from China

Source	Best For	Payment
1688.com	Wholesale, bulk	Alipay, agent
Taobao	Individual components	Alipay, agent
JD.com	Branded (safer warranties)	International cards
AliExpress	Small orders, intl shipping	Cards, PayPal
Shenzhen markets	Best prices, verify in-person	Cash, WeChat

12. SUPPORTING INFRASTRUCTURE

Critical infrastructure requirements for Nigeria — essential for reliable operations.

Item	Specification	Cost (Nigeria)
UPS	3kVA online UPS (APC/Vertiv)	₦800K-₦1.2M
Solar backup	5kVA hybrid inverter + batteries (use VS Solar!)	₦2M-₦3.5M
Server room AC	2HP inverter split unit (dedicated)	₦650K-₦1M
Rack + cabling	12U-42U rack + network cables	₦300K-₦600K
Internet	100Mbps fiber (MainOne/MTN)	₦100K-₦200K/month
Firewall/router	Mikrotik/Ubiquiti edge	₦150K-₦300K
Total infrastructure		**₦3.9M-₦6.6M one-time + ₦100-200K/month**

Complete Setup Cost Summary (Zero-Waste Approach)

Setup Level	Hardware (Landed)	Infrastructure	Total One-Time	Monthly
Phase 1 (TRX50 + 1 GPU)	₦8.2M-₦11M	₦3.9M-₦6.6M	**₦12.1M-₦17.6M**	₦200-500K
Phase 1 + 2 (add 2nd GPU)	₦12.3M-₦16.3M	₦3.9M-₦6.6M (no additional)	**₦16.2M-₦22.9M**	₦200-500K
Phase 3 (co-locate)	Same hardware	₦100-250K/month (co-lo replaces power costs)	**₦0 additional one-time**	₦300-750K

Recommendation: Start with TRX50 Workstation + cloud GPU for overflow (Phase 1). Add 2nd GPU when revenue justifies (Phase 2). Co-locate for production reliability (Phase 3). **Zero components wasted across all phases.**

13. AI TEAM STRUCTURE & NIGERIAN SALARY RANGES

Salary ranges below reflect the actual Lagos tech market — calibrated in Naira, not US dollar equivalents.

AI Team Roles

Role	Skills Required	Salary (₦/month)	Hiring Phase

AI/ML Engineer	PyTorch, HuggingFace, LangChain, vLLM, CUDA	600,000-1,000,000	Phase 2 (Months 5-8)
AI Knowledge/Data Engineer	SQL, Spark, ETL, Qdrant, embeddings	400,000-600,000	Phase 2 (Months 5-8)
MLOps Engineer	Docker, Kubernetes, vLLM, GPU orchestration	500,000-800,000	Phase 3 (Months 9-14)
API/Platform Engineer	FastAPI, APISIX, microservices, REST/gRPC	450,000-700,000	Phase 2 (Months 5-8)
AI Product Manager	AI product roadmap, user research, metrics	500,000-800,000	Phase 3 (Months 9-14)
Developer Advocate	Developer relations, docs, SDK, community	400,000-600,000	Phase 3 (Months 9-14)
Installer Training Manager	Solar systems, technician coordination	350,000-500,000	Phase 2 (Months 5-8)
Fintech/Agent Network Mgr	Agent network, compliance, payment ops	400,000-600,000	Phase 2 (Months 5-8)
Technical Writer	API docs, user guides, training materials	250,000-400,000	Phase 2 (Months 5-8)
AI/ML Intern	Python, basic ML, eager to learn	80,000-120,000	Phase 1 (Months 1-4)

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14. TOTAL BUDGET BREAKDOWN

AI-Specific Budget (within ₦250M total allocation)

Category	One-Time	Monthly	24-Month Total
Hardware (Starter, landed)	₦7M	—	₦7M
Infrastructure (power, cooling, rack)	₦5M	₦150K	₦8.6M
Cloud GPU (Vast.ai/RunPod)	—	₦300K	₦7.2M
AI Team (4 people avg, Phase 2+)	—	₦2M	₦32M (16 months)
Software licenses	—	—	₦0 (all open-source)
Fine-tuning (cloud GPU)	—	—	₦500K (when needed)
Total AI budget	**₦12M**	**₦2.45M**	**~₦55M**

This fits within the ₦250M total budget (₦28M tech + portion of ₦70M salaries + portion of ₦75M hardware).

15. IMPLEMENTATION ROADMAP

Phase 1: Deploy & Learn (Months 1-6)

Action	Timeline	Cost
Purchase & ship Starter server from China	Month 1-2	₦7M
Set up server room (power, cooling, rack)	Month 2-3	₦5M
Install Ubuntu + CUDA + Docker + Ollama	Month 3	₦0
Deploy Qwen2.5-7B + Phi-3 + Mistral-7B via Ollama	Month 3-4	₦0
Set up Qdrant vector DB + LangChain RAG pipeline	Month 4-5	₦0
Build first AI features (solar sizing, product search)	Month 5-6	₦0
Hire AI/ML Intern	Month 1	₦100K/mo

Phase 2: Integrate & Scale (Months 7-18)

Action	Timeline	Cost
Hire AI/ML Engineer + Knowledge Engineer	Month 5-8	₦1M/mo team
Migrate to vLLM for production serving	Month 7-8	₦0
Deploy accounting/auditing/tax AI modules	Month 8-10	₦0

Deploy marketplace AI (recommendations, pricing)	Month 9-12	₦0
Fine-tune Qwen2.5 on business data (LoRA/QLoRA)	Month 10-12	₦200K cloud GPU
Deploy installer AI + PMAP integration	Month 12-15	₦0
Launch paid AI subscription tiers	Month 12	Revenue starts
Add cloud GPU for Mixtral (fraud detection)	Month 12+	₦300K/mo

Phase 3: Proprietary & Scale (Months 19-24+)

Action	Timeline	Cost
Hire MLOps Engineer + AI PM	Month 9-14	₦1.3M/mo team
Upgrade to Production server cluster	Month 18-20	₦12M
Train proprietary model on PECH data	Month 20+	₦500K-₦2M
Launch developer AI platform (public APIs)	Month 20-22	₦0
Rebrand fine-tuned model as "PECH AI"	Month 22+	₦0

16. LICENSING & LEGAL — PER-MODEL VERDICTS

Models SAFE to Rebrand & Close-Source

Model	License	Verdict
Qwen2.5 (all sizes)	Apache-2.0	SAFE — rebrand, modify, close-source freely
Phi-3 / Phi-4	MIT	SAFE — no restrictions
Mistral-7B / Mixtral-8x7B	Apache-2.0	SAFE — but only these specific models
DeepSeek-Coder-V2	MIT	SAFE — check specific model card for MAU limits
SmolLM	Apache-2.0	SAFE
TinyLlama	Apache-2.0	SAFE
Whisper	MIT	SAFE
Qwen-VL	Apache-2.0	SAFE
H2O Danube	Apache-2.0	SAFE

Models with RESTRICTIONS

Model	License	Issue	Recommendation
Llama 3.x	Llama Community	700M MAU cap, can't train competing model	OK for internal tools, DON'T rebrand
Gemma 2/3	Gemma License	Redistribution terms vary	OK for products, check per-version
Mistral Large / Le Chat	Proprietary	NOT open-source	DO NOT USE
DeepSeek (some variants)	Model License	>10M MAU requires permission	Check specific model card

Models to AVOID (Replaced)

Model	Issue	Replaced With
PCMind-2.1-Kaiyuan	Obscure, no community	Qwen2.5-0.5B
Yuan3.0 Flash	Obscure, limited ecosystem	SmolLM 1.7B
Apertus LLM	Niche, limited adoption	Phi-3 Mini

17. RISK ASSESSMENT — NIGERIA-SPECIFIC

Risk	Likelihood	Impact	Mitigation

Power outages	Very High	High — server crashes, data loss	UPS + solar backup + auto-save
Import delays	High	Medium — delayed deployment	Order early, use air cargo if urgent
FX volatility	High	High — budget overruns	Lock prices when buying, maintain USD reserve
Skills gap	High	Medium — slow development	Remote hires, training program, AI interns
Internet disruption	Medium	Medium — AI services offline	Dual ISP, mobile backup, edge caching
Cooling failure	Medium	High — GPU thermal throttle/damage	Dedicated AC + temperature monitoring
GPU hardware failure	Low	High — service disruption	Cloud GPU fallback, warranty
Model quality issues	Medium	Low — poor AI responses	A/B testing, human review, feedback loops
Regulatory changes	Low	Medium — compliance costs	Monitor NDPA, CBN, NCC updates
Data breach	Low	Very High — reputation, legal	Encryption, access controls, NDPA compliance

This document consolidates and fact-checks two strategic planning conversations (5,748 lines total) into a single actionable reference for PECH Group Holdings Ltd. All prices, licenses, and specifications have been verified for the Nigerian market as of March 2026.

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